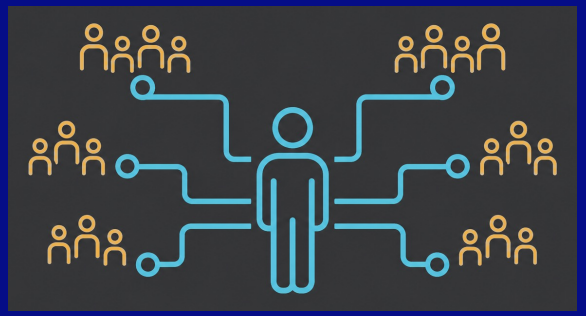


DHUM101
Socially Responsible Computing

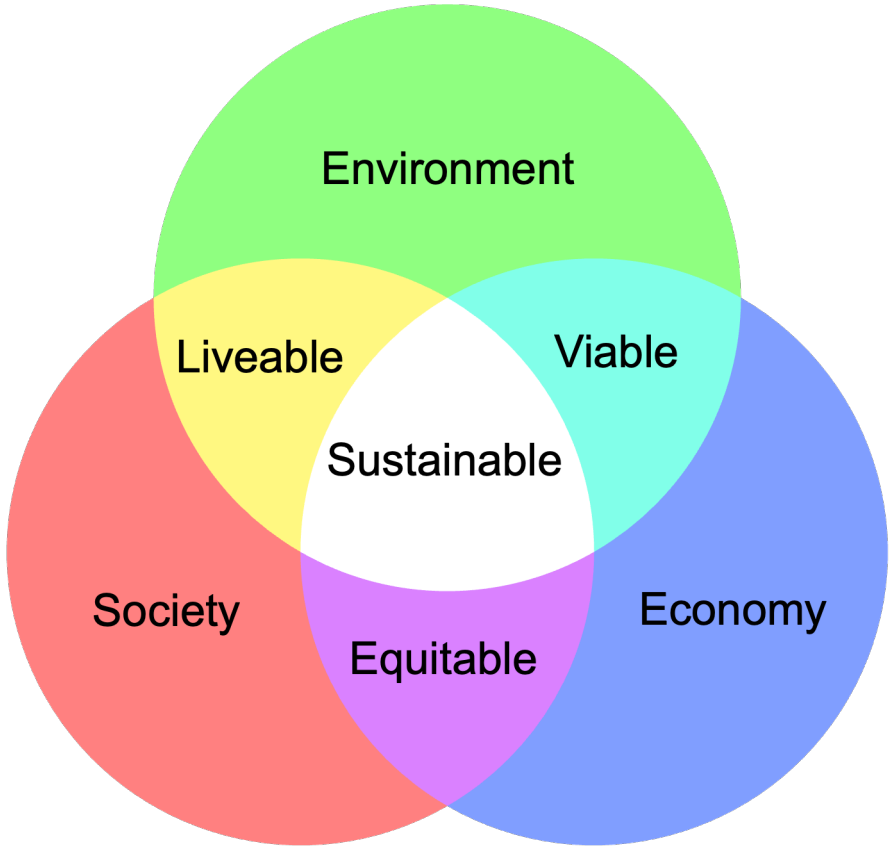
University of Otago
ŌTĀKOU WHAKAIHU WAKA
Te Kura Rorohiko
School of Computing
Te Reo Inarahi me te Mātai Wetereo
English and Linguistics

Lecture 7
Sustainable Computing

- Stephen Cranefield
Semester 2, 2026

What is sustainability?

Sustainable Development





The UN Sustainable Development Goals

Venn diagram translated from https://commons.wikimedia.org/wiki/File:Sch%C3%A9ma_du_d%C3%A9veloppement_durable.svg, by Vigneron, Licence CC BY-SA 3.0

Computing and economic sustainability

An AI Bubble?

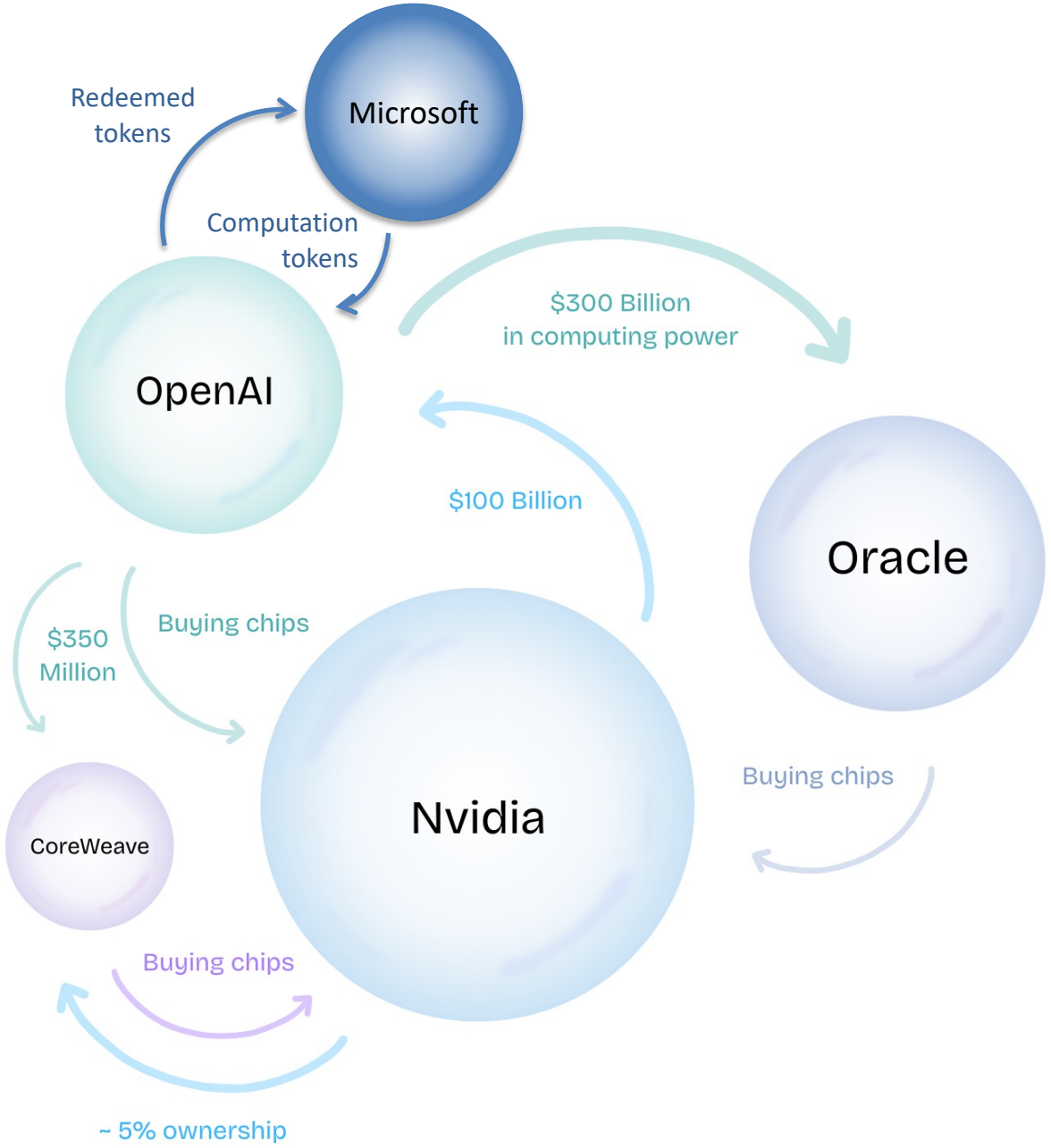
"Even by the standards of tech bubbles, the AI bubble is a whopper. Hundreds of billions of actual dollars have been spent on AI hardware and data centers since OpenAI shipped ChatGPT 3 in 2020.

But even more hundreds of billions of *imaginary dollars* have been spent on AI. For example, Microsoft has a long-standing partnership with OpenAI. As part of this partnership, Microsoft gives OpenAI "tokens" that OpenAI can spend to access the computers in Microsoft's data centers.

OpenAI books these tokens as investment revenue. … When Microsoft "invests" $10 billion worth of tokens in OpenAI, that "$10 billion" figure assumes that if you or I showed up at Microsoft and bought $1 worth of computing, and then Sam Altman showed up and ordered $10 billion worth of computing, Microsoft would simply multiply the amount of computing it sold to us by ten billion [with no discount for buying many tokens].

The really weird math comes after OpenAI redeems its tokens with Microsoft to power ChatGPT and its other products: Microsoft books that transaction as $10 billion worth of AI-related *revenue* to its cloud-computing division."

Corey Doctorow, Chapter "Really weird math", The Reverse Centaur's Guide to Life After AI. Verso, 2026



Bubble diagram modified from https://en.wikipedia.org/wiki/AI_bubble, Licence CC0 (public domain)

What happens when … ?

- "In the first half of 2025, spending on data centers accounted for most of the economic growth in the U.S., outpacing all consumer spending combined".
  https://www.theringer.com/2025/11/04/tech/ai-bubble-burst-popping-explained-collapse-or-not-chatgpt
- "An AI-led market collapse [could] wipe out approximately $33 trillion of value — more than US GDP"
  https://www.oliverwyman.com/our-expertise/insights/2026/jan/impact-ai-bubble-burst-on-global-financial-markets.html
- Some investments funds track an index mechanically, e.g. buying any stock in the Nasdaq-100 without human intervention. Your investments may have high exposure to the performance of big tech companies.
- "A sharp contraction in tech would put tens of thousands of people out of work, vaporize trillions of investment dollars, torpedo retirement and education funds, obliterate life savings, and ruin lives."
  https://www.theringer.com/2025/11/04/tech/ai-bubble-burst-popping-explained-collapse-or-not-chatgpt



https://www.publicdomainpictures.net/en/view-image.php?image=153567&picture=water-balloon-popping.
License: CC0 Public Domain

Computing and Environmental Sustainability (with a focus on data centres)

Cloud computing is efficient at high scale

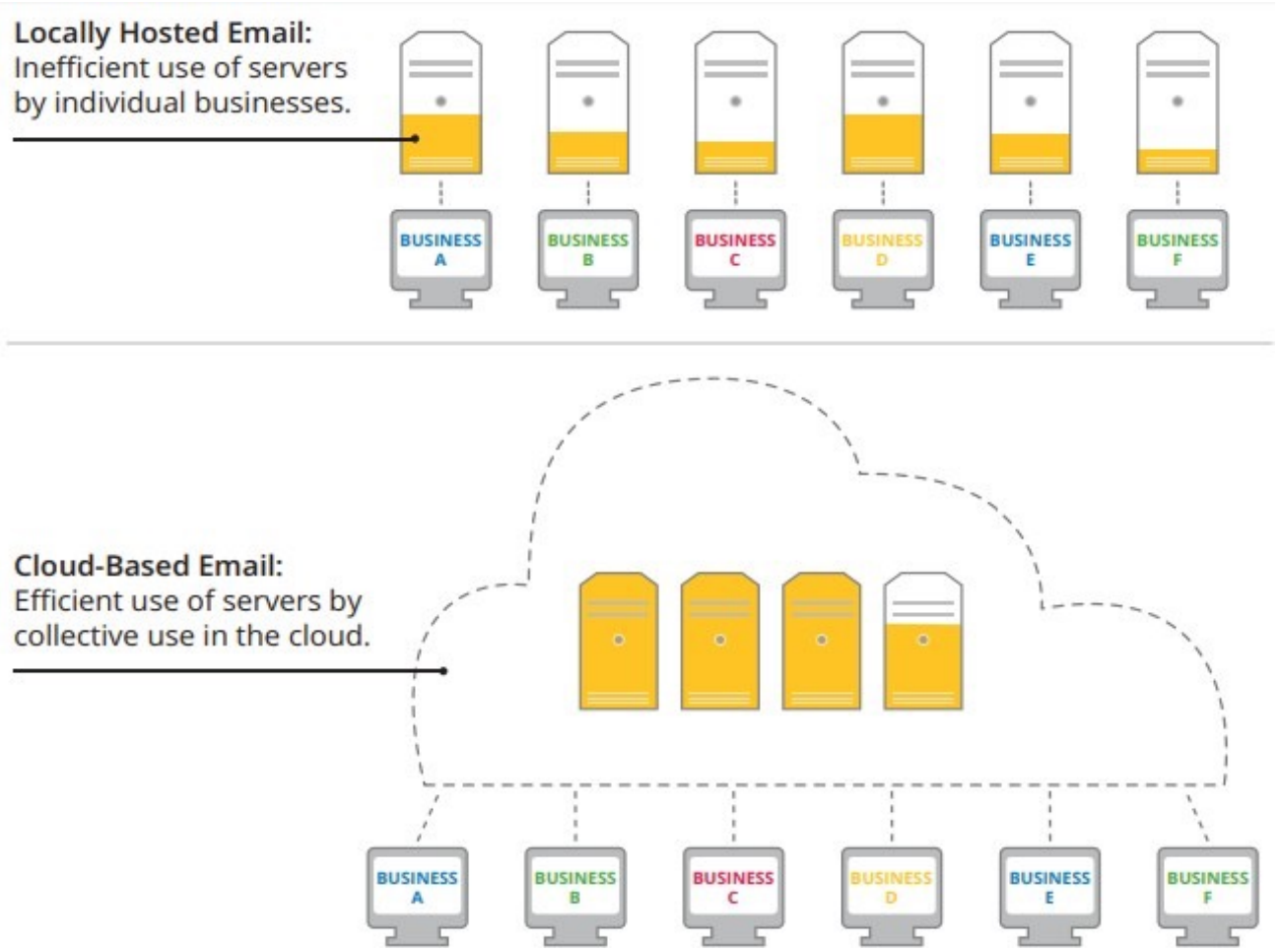


Diagram from Google's Green Computing: Efficiency at Scale (2011), https://www.google.com/green/pdfs/google-green-computing.pdf (Used with permission)

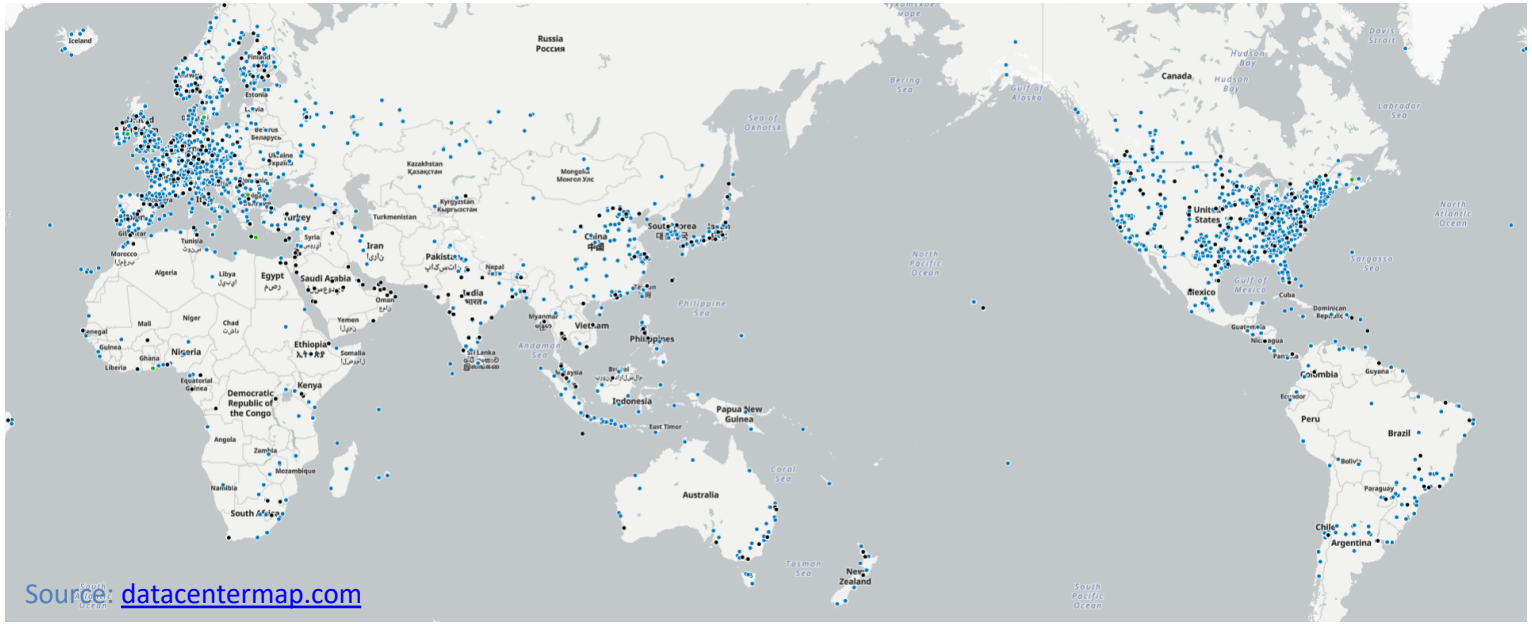
But ...

- Computing needs electricity, which may cause pollution if the electricity grid is not supplied totally from renewable energy sources.
- Computing at scale (in data centres) needs cooling systems, which traditionally uses freshwater, depleting local water sources.
- Data centres can also have *social* impacts on the communities where they are built.
- These are all examples of *externalities*:
  - costs or benefits that arise from an economic activity and affects a third party who is not directly involved in that activity.
- While a company hosting its own computing can (if it chooses) monitor and track these externalities, it is much harder to know what externalities result from cloud computing. The consumer usually just sees the cost.
- Some companies, e.g. Google, publish environmental reports (example), but the AI race means they are going backwards in their efforts to reduce their carbon footprint (Google's was up 18% in 2025, due to the "the buildout of AI infrastructure").



Data center statistics

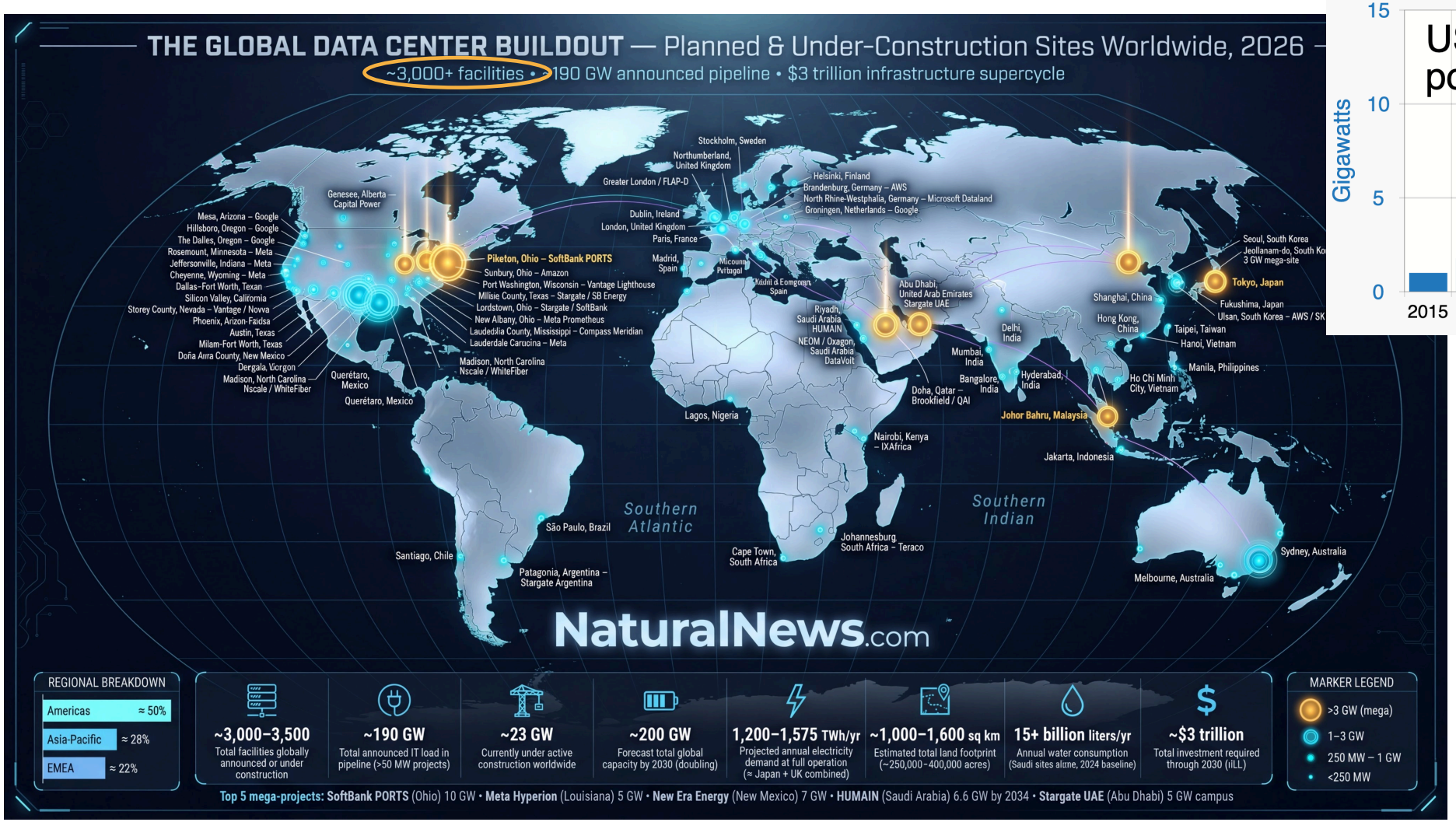
- “There are now estimated to be about 10,000 datacentres worldwide” and “Datacentres are consuming 6% of electricity in the UK and US” [1].
- Microsoft is estimated to have 6.4 million GPUs deployed in data centres [2].

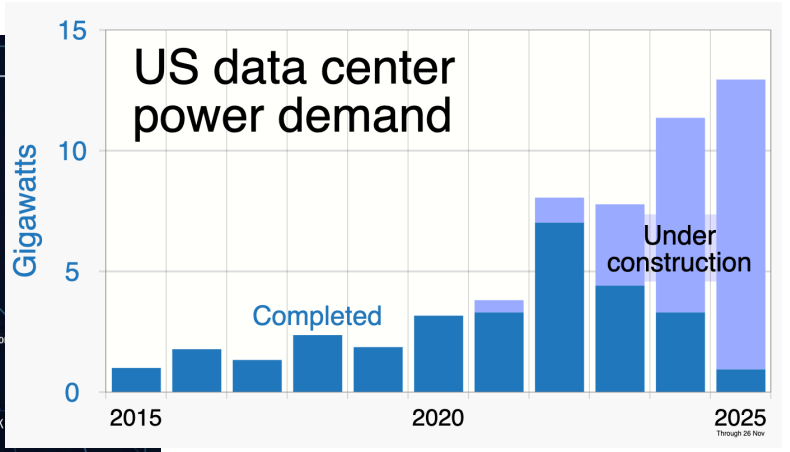


References:

1. https://www.theguardian.com/technology/2026/may/13/datacentres-electricity-consumption-uk-us-ai
2. https://www.theguardian.com/technology/2026/aug/17/are-microsofts-ai-plans-being-held-back-by-a-shortage-of-chips

Data centre proliferation





By RCraig09 - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=180804356

Not shown on the map: there are 20 new data centres planned in NZ

https://technewzealand.org.nz/reports/empowering-aotearoa-new-zealands-digital-future-our-national-data-centre-infrastructure/

https://x.com/HealthRanger/status/2052066691576672706/photo/1 (free to use)

Data centre proliferation and fossil fuels

- “Our datacentres have to run 100% of the time so we can’t even consider using [renewables] as a primary source,” one industry kingpin told the Financial Times last year.” https://www.theguardian.com/commentisfree/2026/aug/09/slough-famous-the-office-datacentres-locals-hub-europe
- “The non-profit Environmental Integrity Project [reviewed] plans for 74 gas-fired power plants that will cater to datacenters across the US. The group estimates that these facilities combined could emit upwards of 660m tons of greenhouse gas pollution per year – equivalent to the entire country of Australia.” https://www.theguardian.com/technology/2026/jul/06/ai-climate-crisis

Local impacts of data centres

Local impacts are covered in Tutorial 6

One is the need for freshwater, traditionally used in cooling systems (data centres generate a lot of heat which will damage computer components if not removed).

Cooling technology is changing to “closed-loop, chip-level cooling systems” and “submerging servers in engineered, non-conductive fluids” (https://technologymagazine.com/news/how-are-companies-pioneering-data-centre-zero-water-cooling)

But there is also early evidence that data centres can raise temperatures nearby.

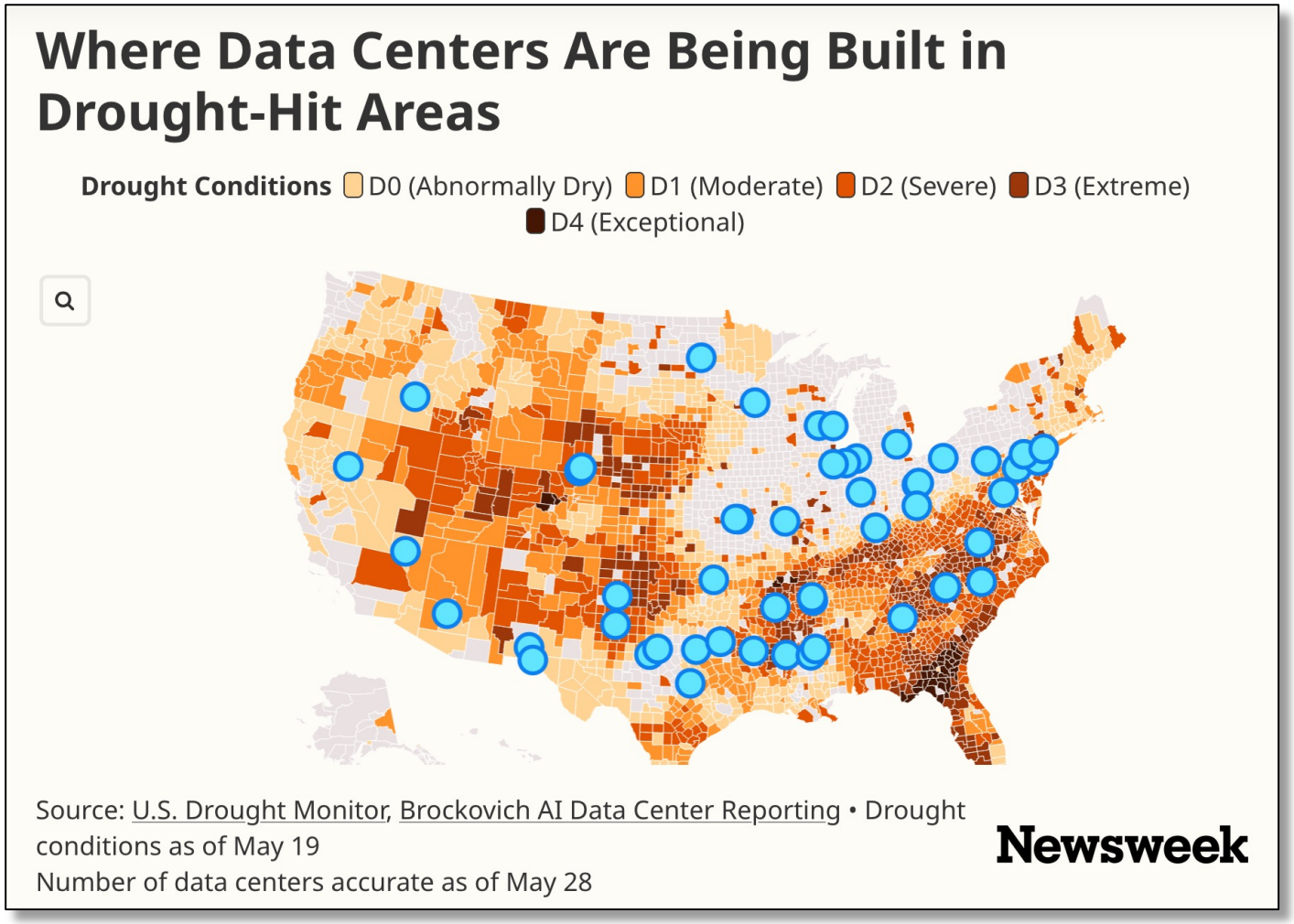


Image Source: https://www.newsweek.com/map-data-centers-built-drought-hit-areas-11997520.
Copied by University of Otago, 23/07/2026

The data centre pushback

Data centre moratoriums

- Called for by more than **230 environmental groups in the US** (Dec. 2025) – see next slide https://www.theguardian.com/technology/2026/feb/23/sam-altman-openai-energy-use-datacenters
- **Seattle** [home to Microsoft and Amazon] has a one-year moratium. https://www.theguardian.com/us-news/2026/jun/09/seattle-ai-datacenters-ban
- **London City Council** has a two-year moratorium. https://www.fuseenergy.com/news/london-data-centre-moratorium
- **The Scottish government** is considering a moratorium due to the 24 "hyperscale" datacentre projects in various stages of planning in Scotland https://www.theguardian.com/uk-news/2026/jul/07/scotland-could-freeze-datacentre-projects-in-challenge-to-uks-ai-strategy

Democratic action

- "In Festus, Missouri, located in a county where Donald Trump won 67% of the vote in 2024, residents filed a petition to recall the mayor and three council members over their approval of a $6bn datacenter agreement with CRG Clayco." https://www.theguardian.com/us-news/2026/jul/03/datacenter-recall-elections
- Concerns about new data centres "have driven thousands of [US] citizens to pack town halls deep into the night to voice their opposition". https://harpers.org/archive/2026/06/the-data-center-divide-andrew-cockburn-artificial-intelligence/

Government policy

- The Australian government published its "Expectations of data centres and AI infrastructure developers" in March 2026 https://www.industry.gov.au/publications/expectations-data-centres-and-ai-infrastructure-developers

The data centre pushback

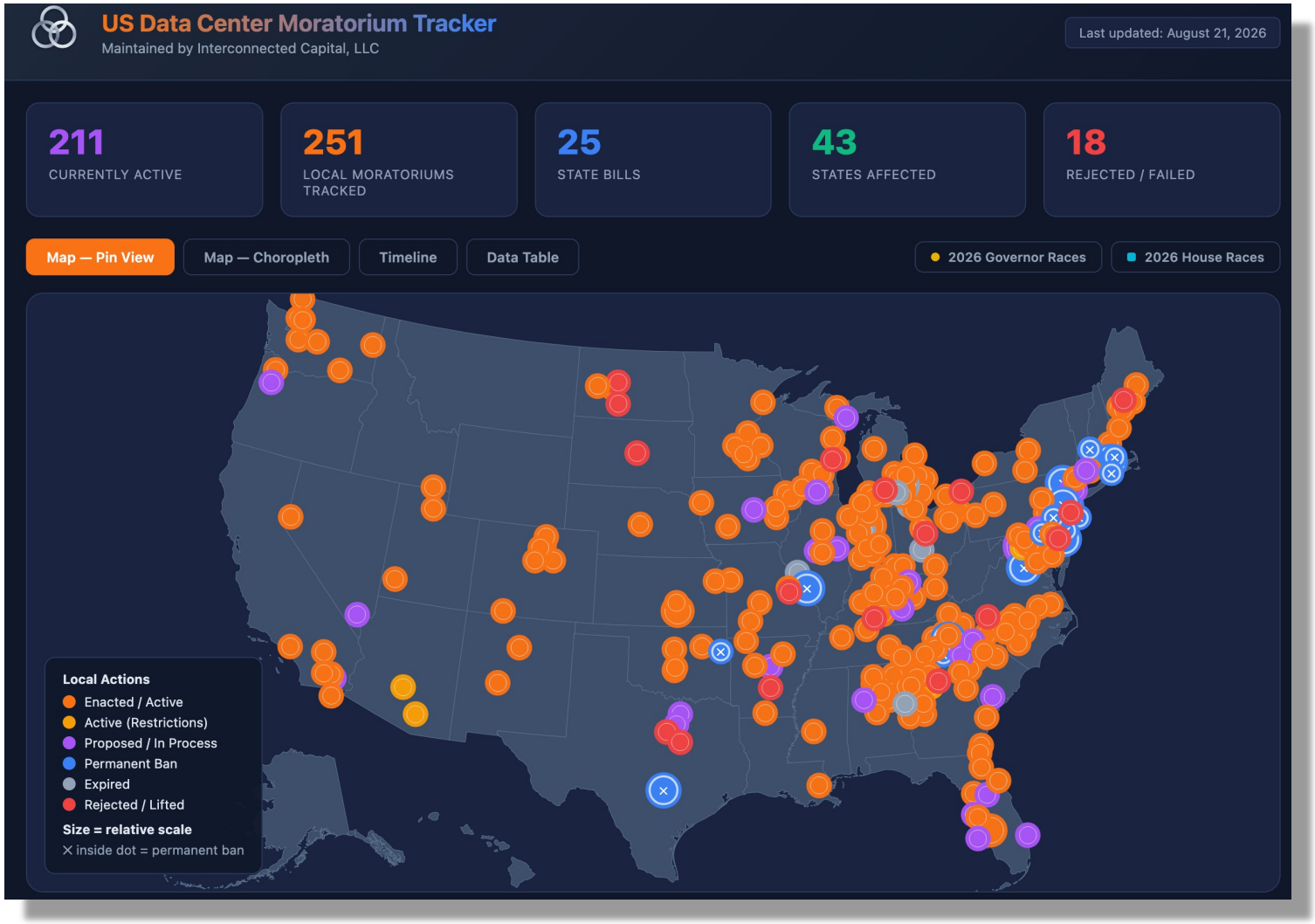
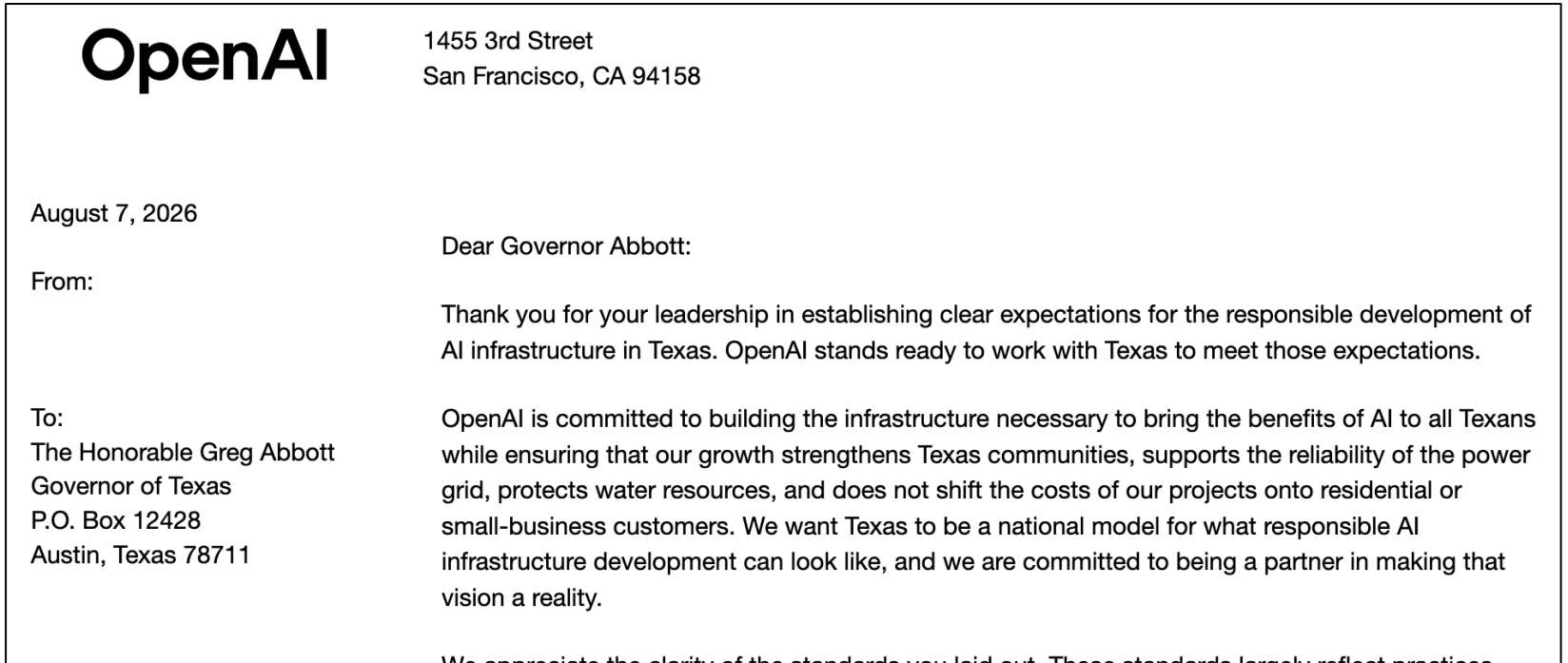




Image Source:
https://www.interconnectedcapital.com/research/data-center-moratoriums
Copied by University of Otago, 23/07/2026

The pushback is working

- “OpenAI sent a letter to Texas Governor Greg Abbott outlining our commitment to responsible AI infrastructure development in Texas. We look forward to working with state and local leaders, utilities, and communities to ensure AI infrastructure delivers meaningful benefits to Texans.” https://openai.com/index/responsible-ai-infrastructure-texas



OpenAI

1455 3rd Street
San Francisco, CA 94158

August 7, 2026

From:

To:
The Honorable Greg Abbott
Governor of Texas
P.O. Box 12428
Austin, Texas 78711

Dear Governor Abbott:

Thank you for your leadership in establishing clear expectations for the responsible development of AI infrastructure in Texas. OpenAI stands ready to work with Texas to meet those expectations.

OpenAI is committed to building the infrastructure necessary to bring the benefits of AI to all Texans while ensuring that our growth strengthens Texas communities, supports the reliability of the power grid, protects water resources, and does not shift the costs of our projects onto residential or small-business customers. We want Texas to be a national model for what responsible AI infrastructure development can look like, and we are committed to being a partner in making that vision a reality.

We appreciate the clarity of the standards you laid out. These standards largely reflect practices

Computing and social sustainability

Social sustainability

Social sustainability is about managing a business's (or a community's) long-term well-being for people. It focuses on impacts on people such as social equity, human rights, decent work, and how companies relate to employees, workers in the value chain, customers, and local communities.

(AI search summary)

Diagram by DALL-E



Social media and social sustainability

- Social media can help to form and sustain communities through information sharing.
- But as information flow on social networks has become more algorithm-driven, many harms have become apparent:
  - Social media addiction, especially in children
    (a US federal trial of Meta brought by 29 state attorneys is currently underway)
  - Appearance-drive content ranked highly by algorithms, social comparisons and peer feedback reinforce harmful beliefs about body image.
  - Increasing polarisation of opinion (especially political) through:
    - "Filter bubbles": when algorithms personalize content for users, leading to exposure only to information that aligns with their existing beliefs.
    - "Echo chambers": when individuals interact primarily with like-minded people, reinforcing their views and isolating them from opposing perspectives.

Source: https://getstream.io/blog/social-media-harms/

Social media content moderators

"Content moderators are the invisible cogs in the vast machinery of Big Tech. There are tens of thousands of them, based all over the world, serving some of the world's best-known tech platforms: Facebook, Instagram, TikTok, Tinder. Their job is to look at images, video and text that has been posted online to decide whether it should be deleted or restricted. Much of what they see is mundane. But some of it leaves lasting scars. ...

Workers have described symptoms of depression, PTSD and anxiety; told us of disturbed sleep, tremors and heightened sensitivity to noise. In multiple cases, moderators told us they had attempted to take their own life.

Others have told me about other, more subtle changes. I've been struck by the number who described a feeling of numbness or desensitisation to disturbing images after the initial horror wore off. One of the more surprising findings from my interviews was the number of moderators who'd taken up smoking."

https://www.thebureauinvestigates.com/stories/2025-05-09/my-two-years-reporting-on-big-techs-hidden-scandal

The AI supply chain

"... we recognize that AI is not just code and algorithms but a vast network of extraction of resources and labor, energy consumption, and waste accumulation—all of which disproportionately affect communities in the so-called Majority World." (https://aiplanetaryjustice.com/ai-supply-chain-impact-framework)

- Raw material extraction
- Materials manufacturing
- Equipment manufacturing
- Model training
- Model deployment
- Disposal/End of Life

The AI supply chain

Raw material extraction → Materials manufacturing → Equipment manufacturing → Model training → Model deployment → Disposal/End of Life

- "I recently returned from the Democratic Republic of Congo, where I saw the reality of artisanal cobalt mining with my own eyes. … estimates suggest that tens of thousands of children work in the mines. … The work is hard, exhausting, and life-threatening. You see people, men and women, breaking the ground with basic tools to extract the minerals from the soil that power our digital world. But my biggest realization was not just the extreme hardship I saw. It was the realization that for countless families in the region, mining is their only economic means of survival."
  https://www.fairphone.com/stories/corporate-distancing-will-not-save-children-from-the-mines
- "As demand for cobalt, gold and other minerals grows, mining is accelerating deforestation in the Congo basin – and increasing the risk of deadly Ebola outbreaks".
  https://www.theguardian.com/world/2026/jun/05/ebola-mineral-mining-smartphones-congo
- "Silver Peak, Nevada [is]the only functioning lithium mine in the United States. … There, batteries for phones, cars, laptops, power banks, surveillance and alarm systems become attainable. … People are forced to leave their land, entire ecosystems are destroyed, and workers risk their lives … Once a smartphone has been physically produced by extracting rare minerals, … the average smartphone lifespan is a mere 4.7 years."
  https://aiplanetaryjustice.com/publications/the-planetary-costs-of-ai

The AI workforce: Data labellers

Raw material extraction → Materials manufacturing → Equipment manufacturing → Model training → Model deployment → Disposal/End of Life (https://aiplanetaryjustice.com/ai-supply-chain-impact-framework)

Data labellers

- "One way companies involved in the AI supply chain try to reduce costs is by employing large numbers of data labellers in countries in the Global South such as the Philippines, Venezuela, Kenya and India. Workers in these countries face stagnating or shrinking wages. For example, an hourly rate for AI data labellers in Venezuela ranges from between 90 cents and US$2. In comparison, in the United States, this rate is between US$10 to US$25 per hour.
- The mental toll of data labelling work is also significant, with repetitive tasks, strict deadlines and rigid quality controls. Data labellers are also sometimes asked to read and label hate speech or other abusive language or material, which has been proven to have negative psychological effects.
  https://theconversation.com/ai-is-a-multi-billion-dollar-industry-its-underpinned-by-an-invisible-and-exploited-workforce-240568

The AI-*enabled* workforce: content reviewers

From https://theconversation.com/like-drinking-from-a-firehose-what-its-like-to-be-the-human-in-the-ai-loop-280708:

Within organisations there are now emerging “content creators” who know how to prompt GenAI tools to quickly generate proposals, reports and presentations even in domains where they lack expertise.

These outputs will be sent to the “content reviewers” for “sanity checks”. Those reviewers are domain experts. They are expected to rectify errors, remove nonfactual hallucinated statements, improve quality and provide accountability and final endorsement. …

On one hand, a GenAI-powered “creator” can generate a plausible 50-page report in a matter of 15–30 minutes. On the other, the “reviewer” will have to spend hours reading, rectifying and rewriting to make the final report ready for the audience.

At one time a creator would be responsible for around 80% of the total time and effort to produce an advanced draft or prototype, and a reviewer would use the remaining 20% to polish it.

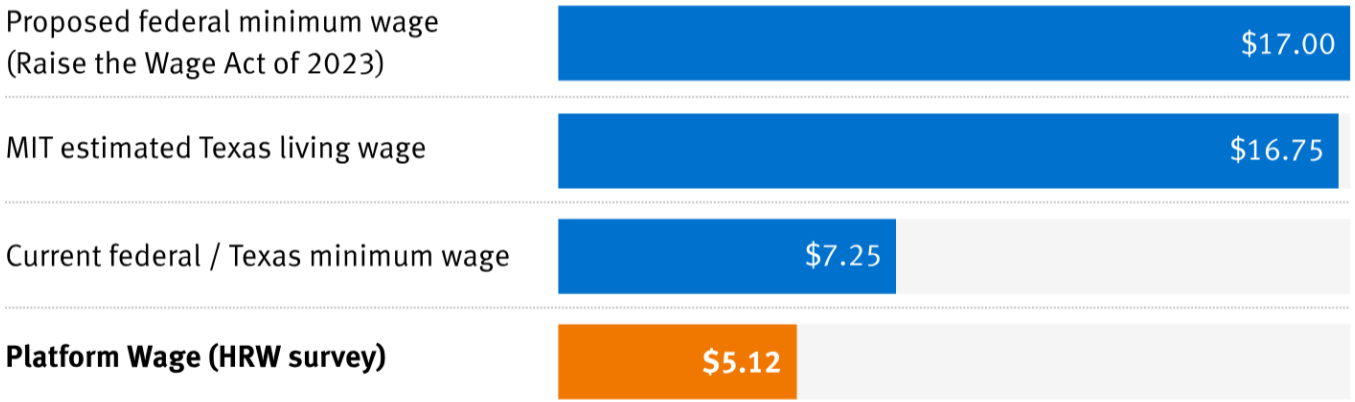
Now the distribution is less than 20% required from the creator, and more than 80% from the reviewer. One of our panellists described this as like “drinking from a firehose”. …

There is …. a personal cost. Subject-matter experts exposed to unrealistic expectations suffer from burnout, low job satisfaction and high turnover in the organisations we spoke with.

The AI-*enabled* workforce: Platform/gig workers

Hourly Wages in Texas

Comparison of hourly wages for platform workers Human Rights Watch surveyed in Texas against other wage metrics

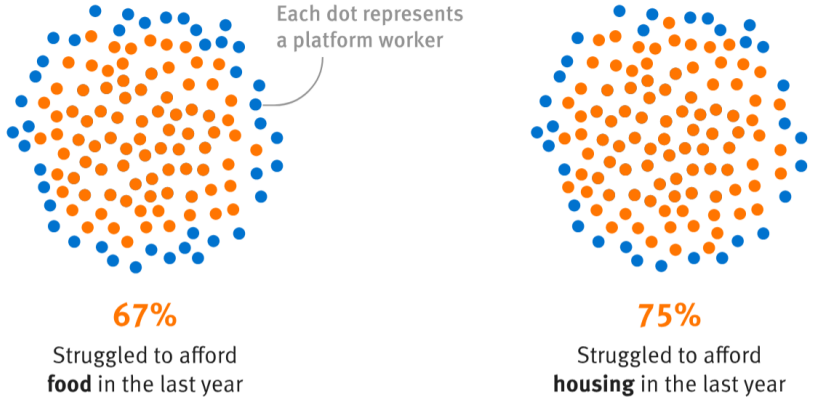


Source: S. 2488 Raise the Wage Act of 2023, MIT Living Wage Calculator, US Department of Labor, Human Rights Watch survey (in order of data points presented).

© 2025 Human Rights Watch

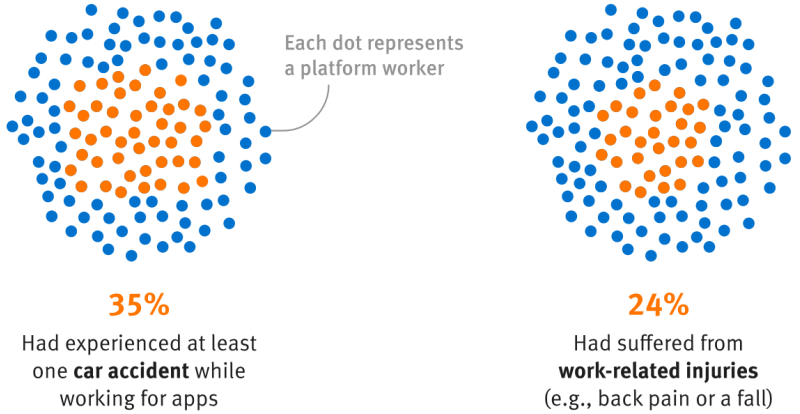
Struggling to Get By

Responses to a Human Rights Watch survey of 127 platform workers in Texas



Threats on the Road and in Their Vehicles

Responses to a Human Rights Watch survey of 127 platform workers in Texas



Source: https://www.hrw.org/report/2025/05/12/the-gig-trap/algorithmic-wage-and-labor-exploitation-in-platform-work-in-the-us,
Licence: CC BY-NC-ND 3.0 US

What can you do?

Learn about and support Green Computing

The study and practice of environmentally sustainable computing, including:

1. optimising energy efficiency during the product's lifecycle;
2. improving the reusability, maintainability, and repairability of the product to extend its lifecycle
3. improving the recyclability or biodegradability of e-waste to support circular economy ambitions; and
4. aligning the manufacture and use of IT systems with environmental and social goals.

Source: https://en.wikipedia.org/wiki/Green_computing





Notable examples of 2 above:

- **Framework computers** – available in NZ (https://frame.work/nz/en)
  "Every computer we ship comes with the only tool you need to replace any part of it in the box. We offer both ready-to-go, pre-assembled versions and our DIY Editions where you can bring your own memory, storage, and OS, including your preferred Linux distro"
- **Fairphone**: "Keep it going for years with swappable parts". "100% e-waste neutral across all products by 2026" (https://www.fairphone.com/

Think about your/your company's choices

"AI is quickly becoming an everyday utility, like electricity. But because it feels invisible, we can easily forget every use has a cost.

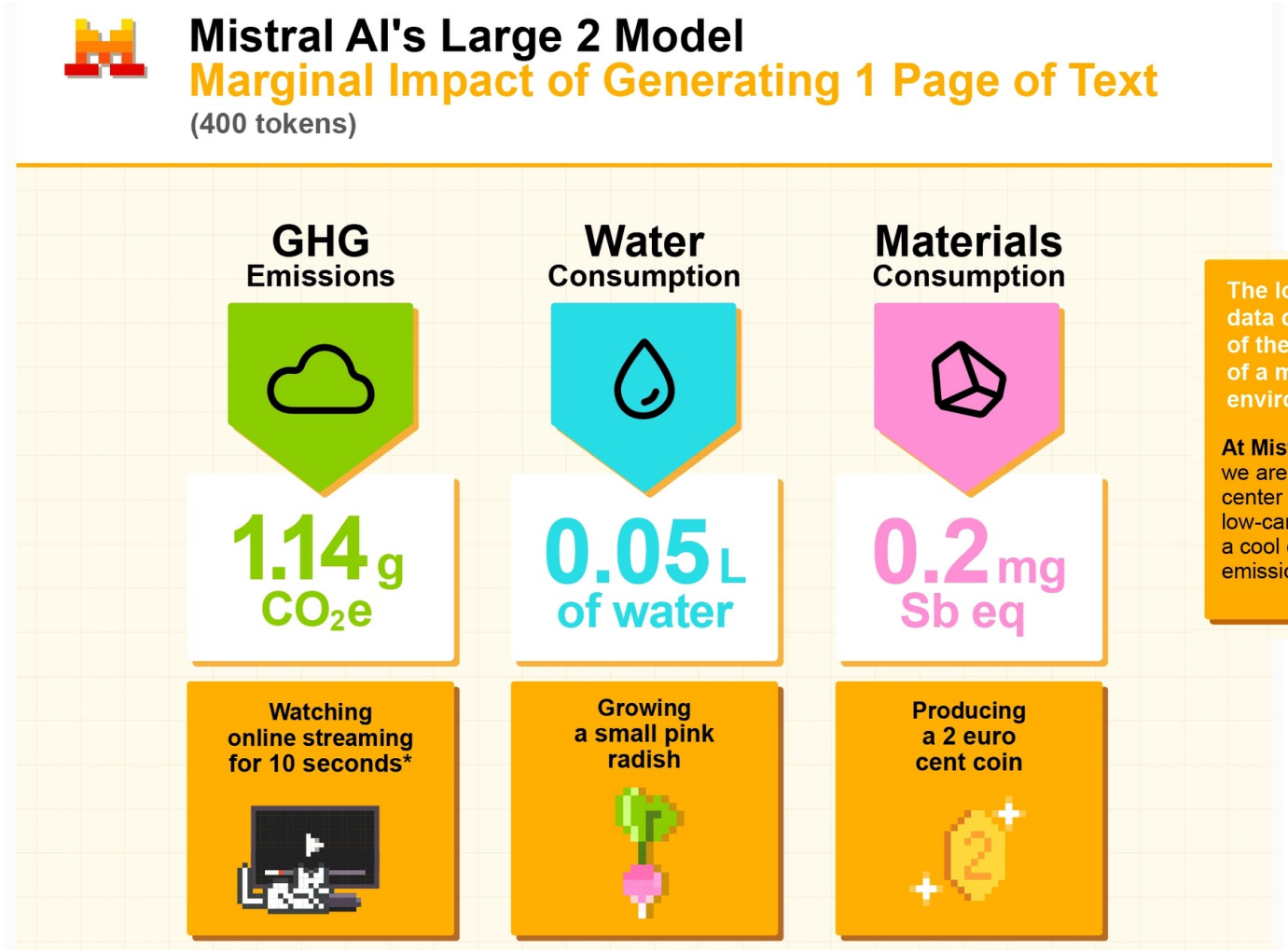
We know not to leave lights on all day. We also know not to run an air conditioner with windows open. We need a similar mindset for AI."

Recommendations:

- "First, choose the right tool for the task. Not every job needs the most powerful AI model. Sometimes a smaller tool, search engine, calculator, or even your own brain is enough. Ask yourself: is AI adding any value here?"
- "Second, write clearer prompts. Vague questions often lead to many follow-up prompts."
- "Third, ask only for what you need. If you need a short list, do not ask for a long report. More words usually mean more computing. "
- "Fourth, be more careful with images, audio, video and other media files. These typically require more computing power than texts.:

https://theconversation.com/your-ai-habit-is-wasting-precious-resources-heres-how-to-use-it-responsibly-285065

Think about your/your company's choices



The location of data centers is one of the main factors of a model's environmental impact.

At Mistral AI,
we are building our own data center in France, leveraging low-carbon electricity and a cool climate to reduce GHG emissions and water usage.

The footprint of a model is strongly correlated with its size. Choosing smaller or case-specific models helps mitigate the environmental impact.

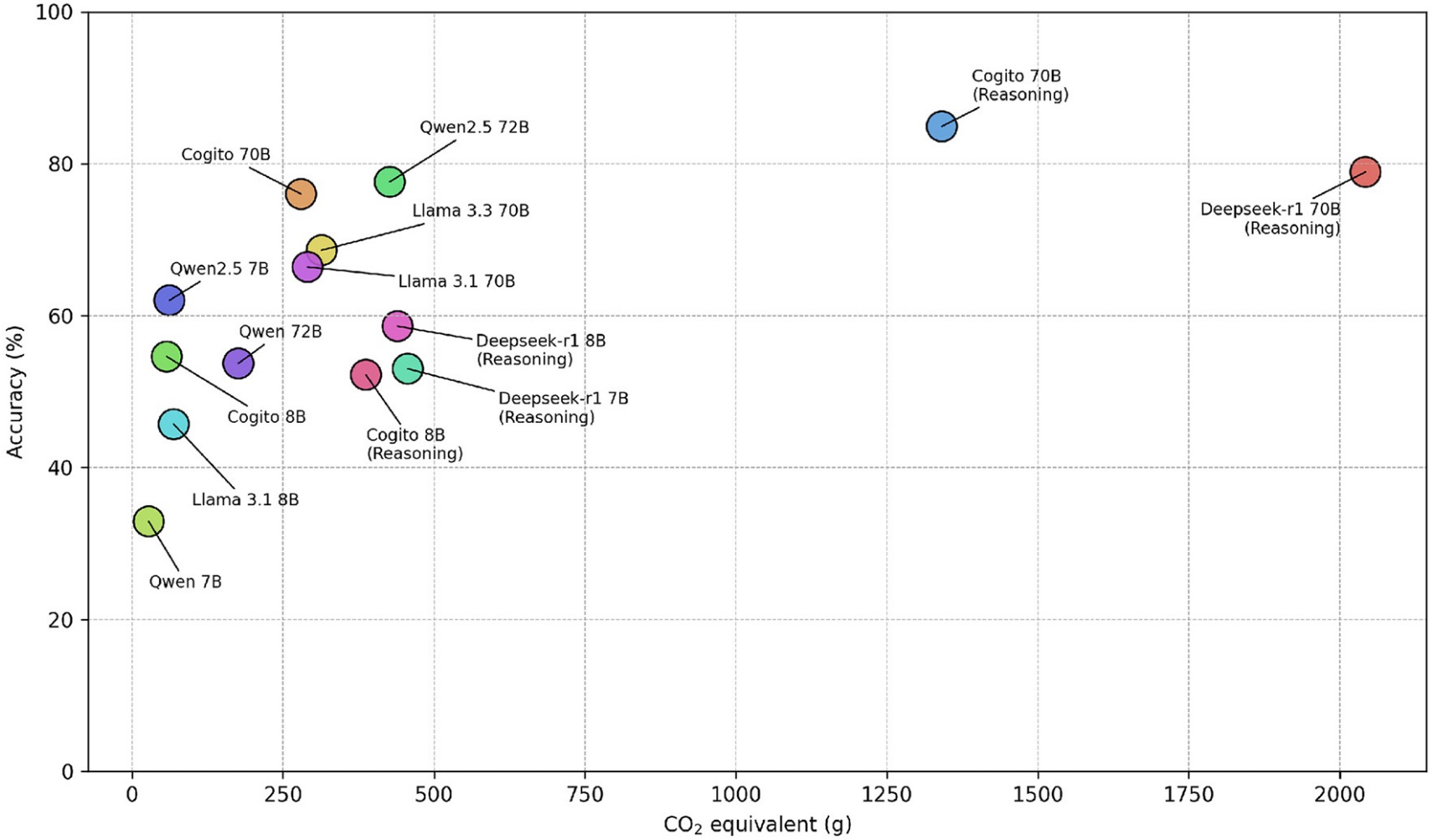
At Mistral AI,
we offer our customers a broad range of model sizes, starting with our smallest model, Ministral 3B.

Inference is a key part of a model's life cycle. End users can also help reducing the impact.

At Mistral AI,
we recommend writing precise prompts and asking for short, grouped answers whenever possible.

https://mistral.ai/news/our-contribution-to-a-global-environmental-standard-for-ai/

Think about your/your company's choices



Combined CO2 eq. emissions (measured in grams) and overall accuracy of each LLM across 1,000 questions. The results illustrate trade-offs between model size, reasoning depth, and environmental impact.

Source:
Dauner M and Socher G (2025) Energy costs of communicating with AI. Front. Commun. https://doi.org/10.3389/fcomm.2025.1572947

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Think about your/your company's choices

- "I don't want to make you feel guilty about sending an email …. I want to make you feel guilty about hosting web infrastructure in Australia.* I want you to see the cloud as computers somewhere, and to ask where those computers are, and to look up what the electricity mix is like where they are. I want you to consider the energy cost of your digital footprint, and count that cost, and report on it publicly. I want you to talk about these issues; bring it up in conversations at work and join a network like the Sustainable Business Council or climateaction.tech."
- "Organisations have the power to change industry standards through their relationships with suppliers, customers and competitors"
- "Internal pressure [from workers in a business] can be powerful"

(Merrin Macleod, Ch. 3 "The environmental impacts of globalised computing", Shouting Zeros and Ones, BWB Texts, 2020 – ebook in the university library)

*Macleod reported (in 2020) that in Australia, datacentres continuously used large amounts of electricity from the New South Wales grid, which is mostly powered by coal.